

```
%Name : Bhumika Vinod Angwani
%PRN : 125B1C042
%Title : Double Integration
%Batch : G2
clc;
clear;
syms x y z;
f=input("Enter the funct  ", "s")
```

```
f =
'(x*exp(y^2))'
```

```
f=str2sym(f)
```

```
f = x ey2
```

```
n=input("Enter no of variable: ")
```

```
n =
2
```

```
for i=1:n
    X=input("Enter the variable name: ")
    ll=input("Enter the ll: ", "s")
    ul=input("Enter the ul: ", "s")
    ll=str2sym(ll)
    ul=str2sym(ul)
    f=int(f,X,ll,ul)
end
```

```
X = y
ll =
'0'
ul =
'x^(1/2)'
```

```
ll = 0
```

```
ul =  $\sqrt{x}$ 
```

```
f =
```

```

$$\frac{x \sqrt{\pi} \operatorname{erfi}(\sqrt{x})}{2}$$

```

```
X = x
```

```
ll =
```

```
'0'
```

```
ul =
```

```
'1'
```

```
ll = 0
```

```
ul = 1
```

```
f =
```

```

$$\frac{e}{8} + \frac{\sqrt{\pi} \operatorname{erfi}(1)}{4} + \frac{3 \sqrt{\pi} \operatorname{erf}(i) i}{16}$$

```

```
disp(f)
```

$$\frac{e}{8} + \frac{\sqrt{\pi} \operatorname{erfi}(1)}{4} + \frac{3 \sqrt{\pi} \operatorname{erf}(i) i}{16}$$

```
disp(simplify(f))
```

$$\frac{e}{8} - \frac{\sqrt{\pi} \operatorname{erf}(i) i}{16}$$